

3: [P1 | P2 | ~~P3~~]
 9: [P1 | P2 | ~~P3~~]
 4: [P1 | ~~P2~~]

8) Time quantum = 10.

9) Apply all CPU scheduling algorithms for insert.

Process	AT	BT	Priority
P1	0	4	1
P2	0	9	2
P3	0	6	3
P4	0	4	1
P5	0	9	2
P6	0	6	3
P7	1	4	1
P8	1	9	2
P9	2	6	3

FCFS:

0: [P1 | P2 | P3 | P4 | P5 | P6]

1: [P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8]

2: P₁ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉

4: ~~P₁~~ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉

13: ~~P₁~~ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉

19: P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉

23: ~~P₃~~ | P₅ | P₆ | P₇ | P₈ | P₉

32: ~~P₃~~ | P₆ | P₇ | P₈ | P₉

38: ~~P₃~~ | P₇ | P₈ | P₉

42: ~~P₃~~ | P₈ | P₉

51: ~~P₈~~ | P₉

57: ~~P₉~~

Grant chart!

Grantt chart:

P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9
0	4	13	19	23	32	38	42	51

Process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₂	0	9	13	19	4	4
P ₃	0	6	19	23	32	19
P ₄	0	4	23	32	38	23
P ₅	0	9	32	38	37	32
P ₆	0	4	42	41	41	41
P ₇	1	4	51	50	49	49
P ₈	1	6	57	55		
P ₉	2					

Q2) Mobile operating tasks:

Music player.

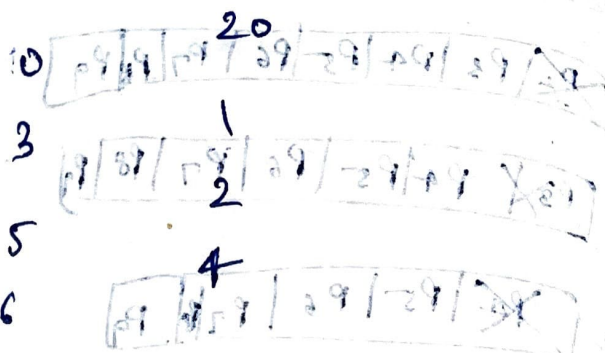
Touch event

Whatsapp notif.

Camera launch.

system has following

Arrival CPU burst



Q3) Which scheduling algorithm is most suitable?

⇒ SR TF (shortest remaining time first) is most suitable.

It is preemptive so newly arriving short task can interrupt long task.

Q4) Execution table:

Music player.

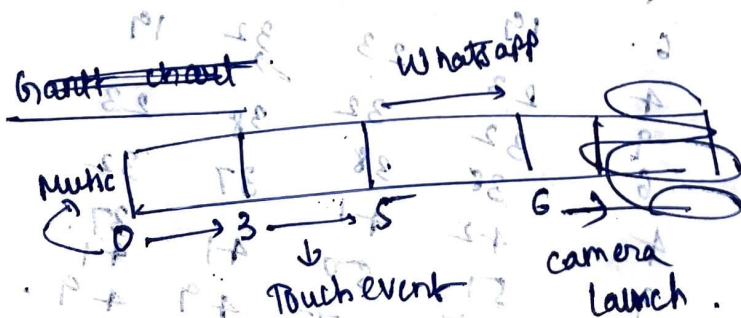
Touch event

Whatsapp notif.

Camera launch.

Arrival

CPU burst



Branch chart

Faster response time:

Music player.

Touch event

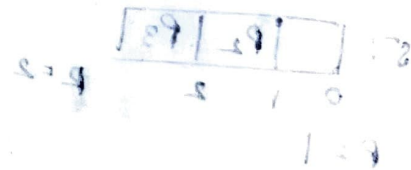
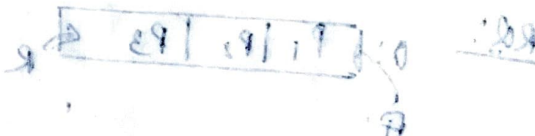
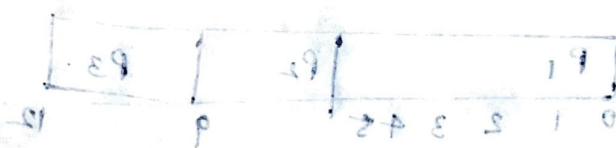
Whatsapp notif

Camera launch.

ne:	PA	PH	1000-1
	2	0	19
Arrival	First start	0	R.T
	8	0	27
0	0		0
3	3		0
5	5		0
6	7		1

1. The first part of the document is a letter from the Secretary of the State of New York to the Governor, dated January 1, 1901. The letter is addressed to the Governor and is signed by the Secretary.

Tests Hand 2737



$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Q3) Consider four processes. priority is 2, 1, 4, 3
 BT is 2, 3, 6, 2.
 Find avg waiting time using priority scheduling algo.

P	BT	Priority
P ₁	2	2
P ₂	3	1
P ₃	6	4
P ₄	2	3

Execution order $\rightarrow P_2 \rightarrow P_1 \rightarrow P_4 \rightarrow P_3$.

Grant chart:

P ₂	P ₁	(P ₄)	P ₃
0	3	24	26

Waiting time = start time - AT.

Process	CT	S.T	WT	TAT
P ₂	3	0	0	3
P ₁	24	3	3	24
P ₄	26	24	24	26
P ₃	32	26	26	32

$$\text{Avg WT} = \frac{0+3+24+26}{4} = \frac{53}{4}$$

$$\text{Avg WT} = 13.25 \text{ units.}$$

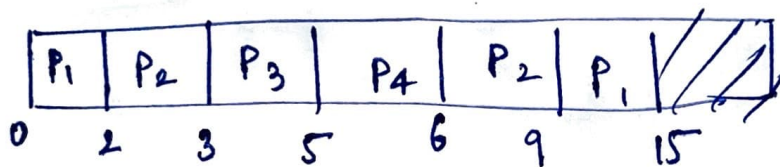
$$\text{Avg TAT} = 21.25$$

Q4) . SRTF (shortest rem. time first).

Process	AT	BT.	CT
P ₁	0	8	8
P ₂	2	4	
P ₃	3	2	
P ₄	5	1	

Soln.

Gantt chart:



Process	CT	TAT (CT-AT)	WT (TAT-BT)
P ₁	15	15	7
P ₂	9	7	3
P ₃	5	2	0
P ₄	6	1	0

$$\text{Avg TAT} = 6.25 \text{ units.}$$

$$\text{Avg WT} = \frac{10}{4}.$$

$$= \underline{\underline{2.5 \text{ units.}}}$$